

Finance Book Formula Sheet

Every numbered equation from the textbook chapter (Financial Statements, Cash Flow, and Taxes) plus the tax formulas from Section 2-9, 2-10, and Web Extension 2A, with what each is used for and the MicroDrive worked values. Built for an open-note test.

Ch 2 — Income Statement and Cash Flow Equations (2-1 to 2-5)

EQUATION	WHAT IT IS USED FOR	HOW TO USE IT (MICRODRIVE 2023, \$M)
(2-1) $EBITDA = EBIT + \text{Depreciation (+ Amortization)}$	Earnings before the non-cash charges; a rough measure of financial strength.	$400 + 200 = \$600$.
(2-2) $EBITDA = \text{Sales} - \text{COGS (excl. depr.)} - \text{Other operating expenses}$	Same number from the top of the statement.	$5,000 - 3,900 - 500 = \$600$.
(2-3) $EPS = \text{Net income to common} \div \text{Shares outstanding}$	The "bottom line" per share.	$248 \div 60 = \$4.13$.
(2-4) $\text{Net cash flow} = \text{NI} - \text{Non-cash revenues} + \text{Non-cash expenses}$ (2-5) $\approx \text{NI} + \text{Depreciation \& amortization}$	Older pen-and-paper cash measure.	$248 + 200 = \$448$.
$\text{Net income} = (\text{EBIT} - \text{Interest}) \times (1 - T)$	Backing out NI, or (reverse) interest from NI: $\text{pre-tax} = \text{NI} \div (1 - T)$.	MicroDrive: $(400 - 60) \times 0.75 = 255$; less preferred dividends 7 = 248 to common.
$\text{Retained earnings}_{\text{end}} = \text{RE}_{\text{begin}} + \text{Net income} - \text{Dividends}$	Statement of stockholders' equity.	$1,552 + 248 - 60 = \$1,740$ (a claim, not cash).

Ch 2 — Free Cash Flow: The Five Steps (2-6 to 2-12)

EQUATION	WHAT IT IS USED FOR	HOW TO USE IT (MICRODRIVE, \$M)
(2-6) NOPAT = EBIT × (1 - Tax rate)	Step 1: operating profit as if the firm had no debt and no financial assets.	400 × 0.75 = \$300 (2022: \$330). Never start from net income.
(2-7) NOWC = Operating current assets - Operating current liabilities = (Op. cash + A/R + Inventory) - (A/P + Accruals)	Step 2: working capital funded by investors.	2023: (100 + 500 + 1,000) - (200 + 400) = \$1,000. 2022: \$710. Excludes short-term investments and notes payable ("pays/charges interest → not operating").
(2-8) Total net operating capital = NOWC + Operating long-term assets	Step 3: all capital used to run the business.	2023: 1,000 + 2,000 = \$3,000. 2022: 710 + 1,780 = \$2,490.
(2-9) Investor-supplied capital = Notes payable + LT bonds + Preferred + Common equity (2-10) Investor-supplied operating capital = (2-9) - Short-term investments	Cross-check from the financing side (whole firm only).	150 + 520 + 100 + 2,240 = 3,010; - 10 = \$3,000 ✓ matches (2-8).
Net investment in operating capital = Op. capital _t - Op. capital _{t-1}	Step 4: new capital tied up this year.	3,000 - 2,490 = \$510 (NOWC +41% vs. sales +4.2% → warning).
(2-11) FCF = NOPAT - Net investment in total operating capital	Step 5: cash available to all investors. Drives intrinsic value.	300 - 510 = -\$210.
(2-12) FCF = [EBIT(1 - T) + Depr.] - Gross investment in fixed assets - Investment in NOWC	Financial-press version; gives the same answer (depreciation cancels).	(300 + 200) - 420 - (1,000 - 710) = -\$210.
Uses of FCF = After-tax interest + Net debt repayment + Dividends + Net stock repurchases + Net purchases of ST investments After-tax interest = Interest × (1 - T)	Where the FCF went; must equal FCF from operations.	45 + (-292) + 67 + 0 + (-30) = -\$210 ✓. Issuing securities / selling investments are negative uses.

Ch 2 — Performance Measures (2-13 to 2-19)

EQUATION	WHAT IT IS USED FOR	HOW TO USE IT (MICRODRIVE)
(2-13) $ROIC = NOPAT \div \text{Operating capital}$ Value test: $ROIC > WACC$	Return operations earn on invested capital — the best operating performance measure.	$300 \div 3,000 = 10\%$ vs. WACC 11.5% (2022: 13.25%; industry 13.19%) → value destroyed.
(2-14) $OP = NOPAT \div \text{Sales}$	Operating profit per dollar of sales.	$300 \div 5,000 = 6.0\%$ (was 6.88%; industry 6.75%).
(2-15) $CR = \text{Operating capital} \div \text{Sales}$ $ROIC = OP \div CR$	Capital tied up per dollar of sales — capital utilization.	$3,000 \div 5,000 = 60\%$ (was 51.88%; industry 51.19%). $6\% \div 60\% = 10\%$ ✓.
(2-16) $MVA = (\text{Shares} \times \text{Price}) - \text{Total common equity}$	Cumulative value created for shareholders since inception.	$60 \times \$31 - 2,240 = -\380M . Coca-Cola: $261 - 23 = +\$238\text{B}$.
(2-17) $MVA = \text{Total market value (equity + debt + preferred)} - \text{Total investor-supplied capital}$	Alternative form; book debt usually proxies for market debt.	—
(2-18) $EVA = NOPAT - (\text{Operating capital} \times WACC)$	One year's true economic profit after charging for ALL capital including equity.	$300 - 3,000 \times 0.115 = 300 - 345 = -\45M . Depreciation is NOT added back.
(2-19) $EVA = \text{Operating capital} \times (ROIC - WACC)$	Same thing — shows value is added only when ROIC exceeds WACC.	$3,000 \times (0.10 - 0.115) = -\45M . Self-test: $100 \times (14\% - 10\%) = \4M .

Ch 2 — Corporate Tax Formulas (2-9)

FORMULA	WHAT IT IS USED FOR	HOW TO USE IT
Corporate tax = $21\% \times \text{Taxable income (flat)}$	Federal tax on operating and investment income.	\$85,000 → \$17,850. \$50M → \$10.5M.
Taxable dividends received = $\text{Dividends} \times (1 - 0.50)$ Tax on dividends = $\text{Dividends} \times (1 - \text{Exclusion}) \times T$	50% dividends-received exclusion (softens triple taxation).	\$4M received, 25% rate → $4 \times 0.5 \times 0.25 = \0.5M tax → keep \$3.5M (12.5% effective).
After-tax interest received = $\text{Interest} \times (1 - T)$ After-tax preferred dividend = $\text{Div} \times [1 - (1 - \text{Excl.}) \times T]$	Choosing between bonds and preferred stock for surplus cash.	4.6% bond on \$100M → $4.6 \times 0.75 = \$3.45\text{M}$; 4% preferred → \$3.5M — preferred wins after tax.
Deductible interest $\leq 30\% \times \text{EBIT}$ (excess carries forward)	Interest expense limitation.	EBIT \$10B, interest \$4B → deduct \$3B, carry \$1B.
NOL used = $\text{MIN}(\text{Cumulative prior losses}, 80\% \times \text{Pre-tax income})$ Taxable income = $\text{Pre-tax income} - \text{NOL used}$	Net operating loss carryforward.	Apex: losses \$160M, income \$120M → use \$96M → taxable \$24M → tax (25%) \$6M → \$64M remains.
Pre-tax income needed to pay \$1 of dividends = $\$1 \div (1 - T)$	Why the code favors debt: interest needs \$1, dividends need more.	At 24.8%: $1 \div 0.752 = \$1.33$.
Capital gain = $\text{Sale price} - \text{Purchase price}$ (corporate rate 21%, short- or long-term)	Gains on securities.	T-bill \$980 → \$1,000: \$20 gain, \$4.20 tax. 100 shares \$15 → \$40: \$2,500 gain.

Ch 2 — Personal Tax Formulas (2-10, Web Extension 2A)

FORMULA	WHAT IT IS USED FOR	HOW TO USE IT
Taxable income = Total income – Adjustments (→ AGI) – Standard or itemized deduction	Form 1040 flow.	Jill: \$100,000 – \$12,950 standard deduction = \$87,050.
Tax = Tax on bracket base + Marginal rate × (Taxable income – Base)	Progressive bracket calculation.	Jill: 4,807.50 + 22% × (87,050 – 41,775) = \$14,768. \$35,000: 1,027.50 + 12% × 24,725 = \$3,994.50.
Average tax rate = Total tax ÷ Taxable income Marginal rate = rate on the next dollar	Progressive means average < marginal.	Jill: 14,768 ÷ 87,050 = 17.0% average vs. 22% marginal. Corporations: both 21%.
Payroll tax = 6.2% (SS, capped at \$147,000) + 1.45% (Medicare, uncapped) = 7.65%; employer matches; self-employed pay 15.3%	Take-home pay.	Jill: \$7,650; take-home = 100,000 – 14,768 – 7,650 = \$77,582. A \$1,000 raise nets \$703.50.
After-tax corporate yield = $r \times (1 - T)$ Equivalent corporate yield = Muni yield ÷ (1 – T) Breakeven $T^* = 1 - (\text{Muni} \div \text{Corporate})$	Muni vs. taxable bond decisions.	37% bracket: 5.5% muni ≡ 8.73% corporate. 7.68% bond at 25% → 5.76%. 8% vs. 5.5% → $T^* = 31.25\%$.
LT-Dividends-Gains tax: 0% if taxable income ≤ \$41,675 (single) / \$83,350 (joint); 20% if ≥ \$459,751 / \$496,600; else 15%. Ordinary income fills brackets first.	Qualified dividends and long-term capital gains.	Couple: \$396,600 ordinary + \$150,000 gains → 15% × 100,000 + 20% × 50,000 = \$25,000.
NIIT = 3.8% × MIN(Net investment income, MAGI – threshold) (threshold \$200,000 single / \$250,000 joint)	Extra tax on high earners' investment income.	MAGI \$300,000, investment income \$80,000 → \$3,040; \$150,000 → \$3,800.
Pass-through effective rate = Personal rate × (1 – 0.20) C-corp total rate = 21% + (Dividend rate × 79%)	Comparing business forms.	37% owner: 29.6% (keeps 70.4%) vs. C-corp 21% + 15.8% = 36.8% (keeps 63.2%) — 7.2-point advantage.

Ch 2 — Self-Test Problem Template (Cole Furnaces)

GIVEN	COMPUTE	ANSWER
EBIT \$4M · Depr. \$1M · Interest \$1M · T = 25% · Op. CA \$14M · Op. CL \$4M · Net PP&E \$15M · WACC 10% · Prior op. capital \$24M	$NI = (4 - 1) \times 0.75$	\$2.25M
	Net cash flow = 2.25 + 1	\$3.25M
	$NOPAT = 4 \times 0.75$	\$3.0M
	$NOWC = 14 - 4$; Op. capital = 10 + 15	\$10M; \$25M
	$FCF = 3 - (25 - 24)$	\$2.0M
	$ROIC = 3 \div 25$	12%
	$EVA = 3 - 0.10 \times 25$	\$0.5M